

Integrated Circuit ECE326

Lec01

Course Introduction

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- ☐ **Course Information**
- ☐ **Course Assessment**
- ☐ **Course Content**
- ☐ **Course Long-Term-Vision**
- ☐ **IC Industry In Egypt**
- ☐ **References**

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Course Information



☐ Lecture's Time:

☐ Wednesday

☐ Lecture's Place:

☐

☐ Lectures Materials, Quiz, Assignment, and Project:

☐ Please, check Microsoft team

☐ Contacts:

☐ Email: ARMohamed@eng.zu.edu.eg , Ahmed.reda@sjtu.edu.cn

☐ Whatsapp: 01020202427



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Course Assessment



☐ Final Exam [64%]

- ☐ You must get > **27** of the paper grade or you will **fail** in the course regardless of your course work.

☐ Quiz and Assignment [20%]

- ☐ You must attend all course sessions otherwise you will **lose** some degrees

☐ Midterm [10%]

- ☐ 8th week

☐ Project [6%]

- ☐ upto 10 students



⊘ Forbidden

Copy or cheat during
Exams and even doing the
Quiz and Assignment

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Course Content (1/2)



□ Course Objective

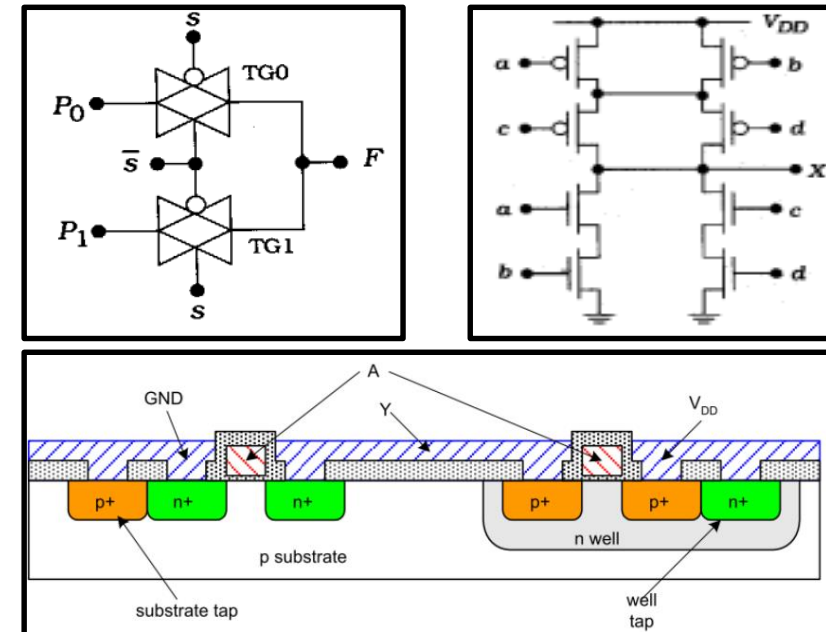
- Overview of important related topics on an integrated circuit and electronic design automation to expand students' horizons about the electronic integrated circuit industry. The course consists of the **three** modules.

□ Module 1: CMOS VLSI Design.

- IC history, design CMOS logic, noise margin, delay time, rising/falling time, Power, **CAD tool**.

□ Module 2: CMOS Processing Technology.

- CMOS technology, Fabrication process...



Course Content (2/2)

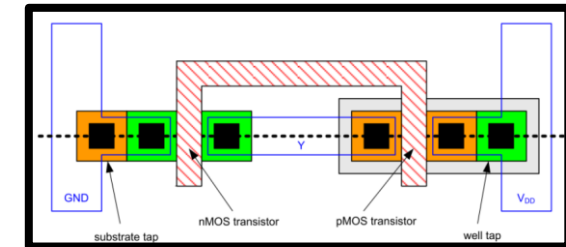


□ Course Objective

- Overview of important related topics on an integrated circuit and electronic design automation to expand students' horizons about the electronic integrated circuit industry. The course consists of the **three** modules.

□ Module 3: IC Mask Design.

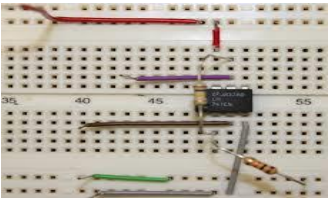
- Digital layout, standard cell technique, parasitic, matching, floor planning, verification, **CAD tool...**



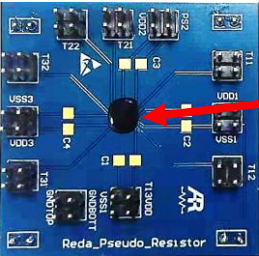
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Course Long-Term-Vision (1/3)

Discrete Element



Bread Board



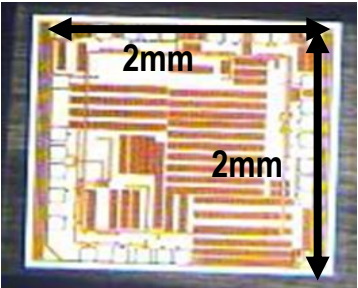
Chip-On-Board (COB)



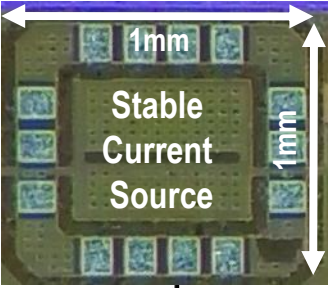
Printed Circuit Board (PCB)

- Through-hole
- Surface mount

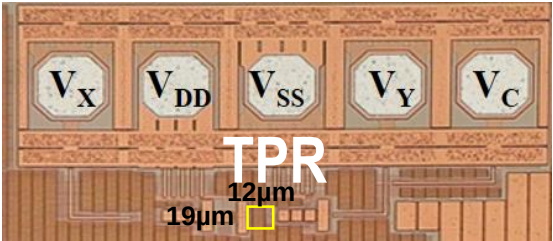
Vey large Scale Integrated Circuit



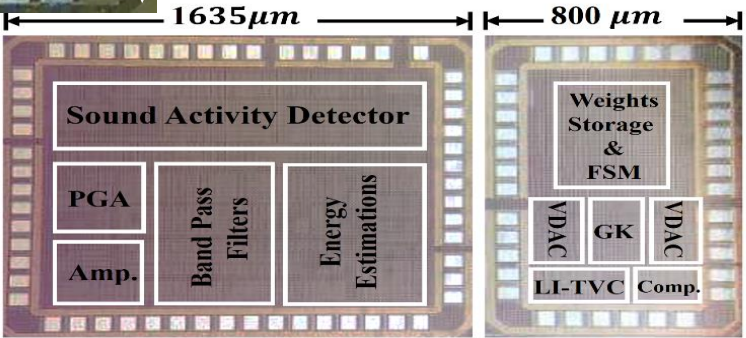
2mm
2mm



1mm
1mm
Stable Current Source

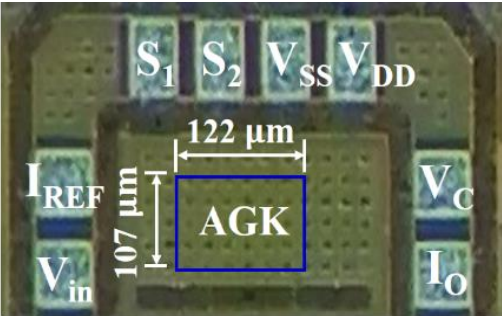


TPR
12µm
19µm



1635µm
800 µm

Sound Activity Detector
PGA
Amp.
Band Pass Filters
Energy Estimations
Weights Storage & FSM
V_{DD}
V_{SS}
V_X
V_Y
V_C
V_{REF}
V_{in}
I_{REF}
I_o
V_C
V_{SS}
S₁
S₂
AGK
LI-TVC
Comp.
V_{DD}
V_{SS}
V_X
V_Y
V_C

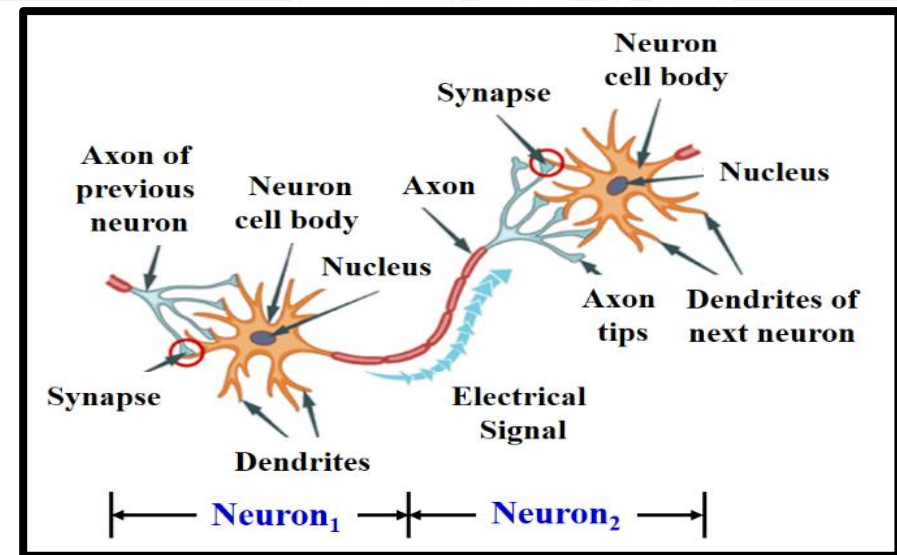
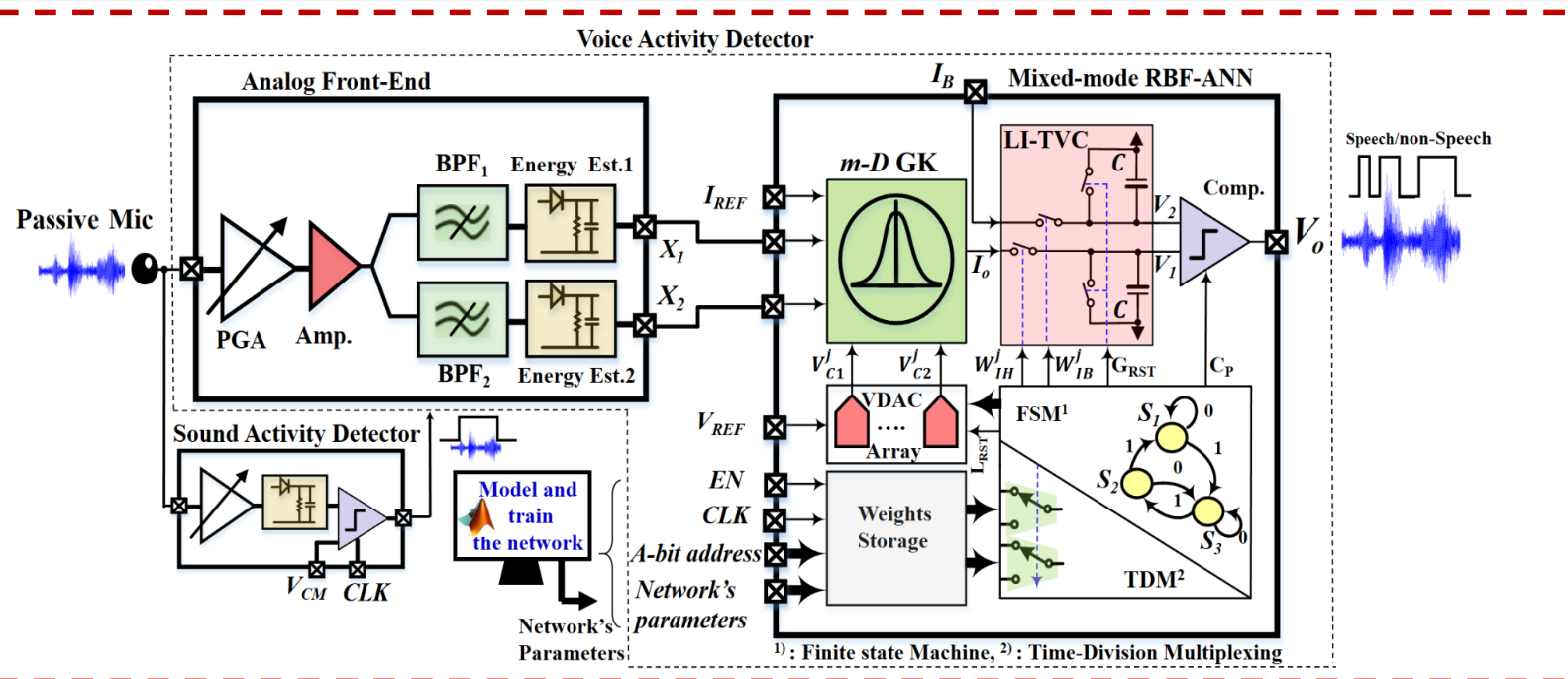


122 µm
107 µm
AGK

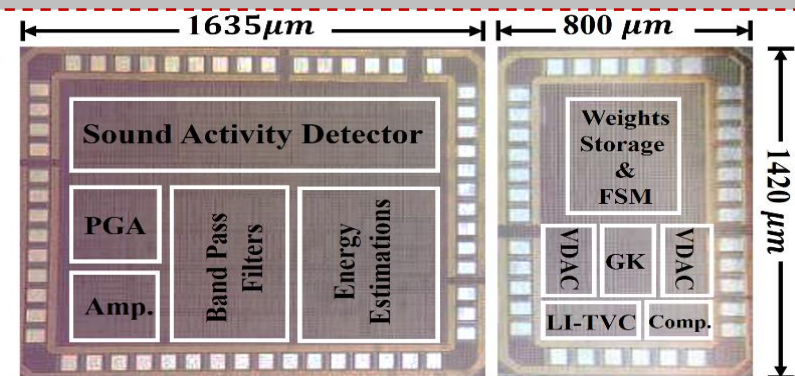
A. Reda

Item	Discrete Element	Very large scale Integrated Circuit
Area	Large	Small
Configuration	Easy	Complex
Price	Cheap	Expensive
Applications	Small production	Mass production
Power consumption	High	Low

Course Long-Term-Vision (2/3)



Brain biological system.



Artificial Neural Network On-Chip





Electronics in Bio-medical Devices

Pacemaker

Cochlear implant

ECG monitoring

ECG	ElectroCardio-Gram
PPG	Photo-Plethy-smo Graphy
EEG	Electro-encephalo- gram

PPG monitoring

EEG monitoring

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IC Industry In Egypt

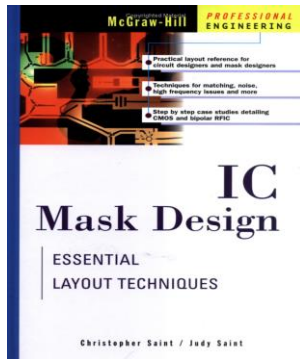
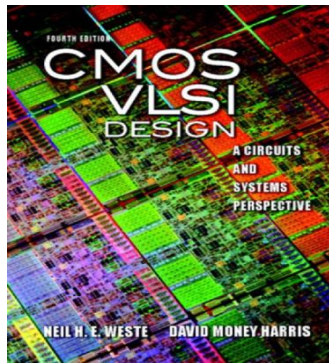
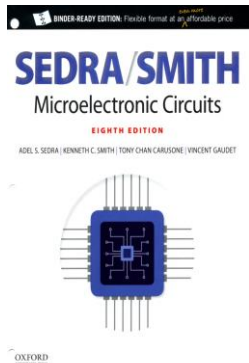


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References



- ❑ A. S. Sedra et. al., **Microelectronic Circuits**, 8th edition, Oxford Series in Electrical and Computer Engineering, 2020
- ❑ N. H. E. Weste et al., **CMOS VLSI Design A Circuits and Systems Perspective**, 4th Addison-Wesley, 2011.
- ❑ C. Saint et. al., **IC Mask Design Essential Layout Techniques**, McGraw-Hill, 2002.



Stay on target



[100%]

- ❑ Lectures
- ❑ Sections
- ❑ Labs
- ❑ Project

Thanks!